QF620 Stochastic Modelling in Finance — Session 7 Reference Cheatsheet

Equivalent Martingale Measure: the Radon–Nikodym derivative, Girsanov's theorem, the road to Black–Scholes, numeraires and martingale pricing, and the risk-neutral Black–Scholes derivation

Built from Session 7 slides (Tee Chyng Wen, SMU) · October 2026 · with symbol-by-symbol equation breakdowns

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Source: "Session 7: Equivalent Martingale Measure" (Tee Chyng Wen, SMU, 19 slides). Slide numbers are PDF page numbers, shown as [S#]. Material not on the slides is tagged *[Beyond slides]*. ⚠marks a slide ambiguity or error, with the corrected version.

How to read the equations: every displayed equation is followed by a shaded **where** box that names each symbol and component and says what it represents in that formula.

Exam context: the exam format is not stated on the slides; check the course outline. From this session you need: the Radon–Nikodym derivative as a ratio of densities and how it changes the measure of an expectation; the statement of Girsanov's theorem and how to use it to change the drift of an SDE; why the option price does not depend on μ; what a numeraire is and why the discounted price is a martingale under the matching measure; and the full EMM derivation of the Black–Scholes call.

Notation used throughout (slide conventions)

Symbol	Meaning	Type	Notes
$\mathbb{P}$	real-world (physical) probability measure	measure	[S4–S5]
$\mathbb{Q}$, $\mathbb{Q}^*$, $\tilde{\mathbb{P}}$	equivalent (martingale) measure; $\mathbb{Q}^*$ = risk-neutral with B_t numeraire	measure	[S5, S13, S17]
$\mathbb{E}^{\mathbb{P}}$, $\mathbb{E}^{\mathbb{Q}}$, $\mathbb{E}^*_t$	expectation under a measure; $\mathbb{E}^*_t$ = risk-neutral, conditional on time-t info	operator	[S4, S13]
$\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}$	Radon–Nikodym derivative (likelihood ratio)	random var. > 0	[S3–S5]
$f(x)$, $\tilde{f}(x)$	density with drift / without drift	function	[S2]
W_t	$\mathbb{P}$-Brownian motion	random	[S5]
W^*_t	$\mathbb{Q}$-Brownian motion, $W^*_t = W_t + \int_0^t \kappa_u \,\mathrm{d}u$	random	[S5, S17]
κ, κ_t	Girsanov kernel (drift shift); in Black–Scholes $\kappa = \frac{\mu - r}{\sigma}$	number / process	[S5, S17]
S_t, B_t	stock; money-market account $B_t = B_0 e^{rt}$	price	[S12, S16]
$X_t = S_t / B_t$	relative (discounted) price	price	[S17]
μ, r, σ	real-world drift, risk-free rate, volatility	number	[S12, S16]
V_t	derivative value; V_T its payoff	price	[S19]
Φ	standard normal CDF	function	[S10, S19]

Notation trap: on [S2–S4] $\mathbb{P}$ is the measure *with* drift $-\kappa$ and $\tilde{\mathbb{P}}$ the driftless one; on [S5] the roles flip — $\mathbb{P}$ is the original measure where W_t is driftless and $\mathbb{Q}$ is the new one under which W_t has drift $-\kappa$. Either way $\frac{\mathrm{d}(\text{drift-}-\kappa\text{ measure})}{\mathrm{d}(\text{driftless measure})} = e^{-\kappa W_t - \frac{1}{2}\kappa^2 t}$. Also κ is a *market price of risk* here, not the Heston mean-reversion speed from Session 6.

0. Topic map

```
PREREQUISITES
S2  binomial risk-neutral q;  S3 Brownian motion, martingales;  S4 Itô;  S5 GBM solution, BS formula
   |
   ▼
SESSION 7: EQUIVALENT MARTINGALE MEASURE
├─ MEASURE CHANGE INTUITION   [S2–S4]
|    ├─ X_t ~ N(−κt, t): density f vs driftless density f̃                    [S2]
|    ├─ Likelihood ratio f/f̃ = exp(−κx − ½κ²t) = Radon–Nikodym derivative     [S3]
|    └─ dP/dP̃ > 0, E_P̃[dP/dP̃] = 1, E_P[g] = E_P̃[g dP/dP̃]; P ~ P̃ equivalent  [S4]
|
├─ GIRSANOV   [S5–S7]
|    ├─ W*_t = W_t + ∫κ du is a Q-BM; dQ/dP = exp(−∫κ dW − ½∫κ² dt)          [S5]
|    ├─ E_P[W]=0, E_Q[W]=−κt, E_Q[W*]=0, E_P[W*]=κt                          [S6]
|    └─ Change drift μ → r in dX = μX dt + σX dW with κ = (μ−r)/σ            [S7]
|
├─ BLACK–SCHOLES HISTORY & ASSUMPTIONS   [S8–S12]
|    ├─ Actuarial pricing: needs μ and a risky discount rate (personal)      [S8, S10]
|    ├─ BS: hedge away stock risk ⇒ only r and σ matter                      [S9–S10]
|    ├─ 1997 Nobel prize (Merton, Scholes; Black)                            [S11]
|    └─ dS = μS dt + σS dW, dB = rB dt; seven assumptions                     [S12]
|
├─ NUMERAIRE & EMM   [S13–S15]
|    ├─ S_t/e^{rt} = E*_t[S_{t+Δt}/e^{r(t+Δt)}]; numeraire = strictly positive  [S13]
|    ├─ Complete market ⇒ hedge ⇒ riskless ⇒ hedged portfolio/B is martingale  [S14]
|    └─ Equivalent measures agree on null sets ⇒ arbitrage is measure-free    [S15]
|
└─ EMM APPLICATION: BLACK–SCHOLES   [S16–S19]
     ├─ μ is the hardest parameter to estimate                               [S16]
     ├─ X = S/B: dX = (μ−r)X dt + σX dW → Girsanov κ=(μ−r)/σ → dX = σX dW*   [S17]
     ├─ dS = rS dt + σS dW*;  S_T = S_0 exp((r−½σ²)T + σW*_T)                [S18]
     └─ V_0/B_0 = E*[V_T/B_T] ⇒ S_0Φ(d₁) − Ke^{−rT}Φ(d₁ − σ√T)               [S19]
   |
   ▼
LATER: S8 static replication (risk-neutral density from option prices); change of numeraire
```

Dependency logic:

- A change of measure is a reweighting of outcomes by a strictly positive density ratio $\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}$. Because the weight is never zero, both measures agree on which events are impossible (they are *equivalent*).
- Girsanov says a specific exponential weight turns a Brownian motion into a Brownian motion with a shifted drift, so it can replace the drift of any Itô SDE without touching the volatility.
- Black–Scholes hedging removes the stock's risk, so the option grows at r and μ drops out. In measure language: pick the numeraire B_t, use Girsanov to make S_t/B_t a martingale, and the option price is the $\mathbb{Q}^*$ expectation of the discounted payoff.

1 Concept entries

1.1 Intuition behind measure change [S2]

A normal process with drift $-\kappa$ has density

$$X_t \sim \mathcal{N}(-\kappa t,\, t), \qquad f(x) = \frac{1}{\sqrt{2\pi t}} \exp\Big[-\frac{(x+\kappa t)^2}{2t} \Big], \qquad \mathbb{E}[g(X_t)] = \int_{-\infty}^{\infty} g(x) f(x)\, \mathrm{d}x.$$

where

X_t	a normal random variable at time t: mean $-\kappa t$, variance t (a Brownian motion with drift $-\kappa$).
κ	drift coefficient; can be positive or negative.
$f(x)$	probability density of X_t under the original measure.
$g(\cdot)$	any bounded function (e.g. a payoff).

Introduce a driftless density and multiply and divide by it inside the integral:

$$\tilde{f}(x) = \frac{1}{\sqrt{2\pi t}} \exp\Big[-\frac{x^2}{2t} \Big], \qquad \mathbb{E}[g(X_t)] = \int_{-\infty}^{\infty} g(x)\, \frac{f(x)}{\tilde{f}(x)}\, \tilde{f}(x)\, \mathrm{d}x.$$

where

$\tilde{f}(x)$	density of a driftless $\mathcal{N}(0,t)$ variable: a standard Brownian motion at time t.
$f/\tilde{f}$	the weight that converts an average under $\tilde{f}$ into an average under f.

1.2 Likelihood ratio = Radon–Nikodym derivative [S3]

Since both densities are non-zero the ratio is well defined:

$$\frac{f(x)}{\tilde{f}(x)} = \frac{\exp\big[-\frac{(x+\kappa t)^2}{2t} \big]}{\exp\big[-\frac{x^2}{2t} \big]} = \exp\Big[-\kappa x - \frac{1}{2}\kappa^2 t \Big].$$

where

$-\kappa x$	from expanding $(x+\kappa t)^2 = x^2 + 2\kappa t x + \kappa^2 t^2$ and dividing by $2t$: the cross term.
$-\frac{1}{2}\kappa^2 t$	the squared-drift term; it makes the ratio average to 1 under $\tilde{f}$.

- This ratio is the **likelihood ratio**, or (in continuous time) the **Radon–Nikodym derivative** [S3].
- Discrete-time model: a ratio of two *probabilities* (e.g. q/p on a binomial tree). Continuous-time model: a ratio of two *probability density functions* [S3].
- Why "derivative": $\int g\, \mathrm{d}\mu = \int g\, \frac{\mathrm{d}\mu}{\mathrm{d}\nu}\, \mathrm{d}\nu$, just like a chain rule for measures [S3].

1.3 Properties of the Radon–Nikodym derivative [S4]

Let $\mathbb{P}$ be the measure with density f and $\tilde{\mathbb{P}}$ the one with density $\tilde{f}$:

$$\frac{\mathrm{d}\mathbb{P}}{\mathrm{d}\tilde{\mathbb{P}}} = \frac{f(x)}{\tilde{f}(x)} > 0, \qquad \mathbb{E}^{\tilde{\mathbb{P}}}\Big[\frac{\mathrm{d}\mathbb{P}}{\mathrm{d}\tilde{\mathbb{P}}} \Big] = 1, \qquad \mathbb{E}^{\mathbb{P}}[g(X_t)] = \mathbb{E}^{\tilde{\mathbb{P}}}\Big[g(X_t) \frac{\mathrm{d}\mathbb{P}}{\mathrm{d}\tilde{\mathbb{P}}} \Big].$$

where

> 0	strictly positive: no outcome is given zero weight.
$\mathbb{E}^{\tilde{\mathbb{P}}}[\cdot] = 1$	total probability is preserved: $\int \frac{f}{\tilde{f}} \tilde{f}\,\mathrm{d}x = \int f\,\mathrm{d}x = 1$.
third identity	the change-of-measure rule: compute a $\mathbb{P}$-expectation by averaging under $\tilde{\mathbb{P}}$ with weights $\frac{\mathrm{d}\mathbb{P}}{\mathrm{d}\tilde{\mathbb{P}}}$.

Slide question [S4]: why are $\mathbb{P}$ and $\tilde{\mathbb{P}}$ equivalent? *[Beyond slides]* Two measures are equivalent when they have the same null sets (they agree on what is possible and impossible [S14]). Both normal densities are strictly positive on all of $\mathbb{R}$, so an event has probability zero under one if and only if it has probability zero under the other. Equivalently, $\frac{\mathrm{d}\mathbb{P}}{\mathrm{d}\tilde{\mathbb{P}}}$ is strictly positive and finite.

1.4 Girsanov's theorem [S5]

Constant-drift version. With $\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}} = e^{-\kappa W_t - \frac{1}{2}\kappa^2 t}$, a $\mathbb{P}$-Brownian motion W_t becomes a Brownian motion with drift $-\kappa$ under $\mathbb{Q}$; that is, $W_t^* = W_t + \kappa t$ is a $\mathbb{Q}$-Brownian motion with

$$\mathbb{E}^{\mathbb{Q}}[W_t^*] = 0, \qquad \mathbb{E}^{\mathbb{Q}}\left[e^{\sigma W_t^*}\right] = e^{\frac{1}{2}\sigma^2 t}, \qquad \mathbb{E}^{\mathbb{Q}}\left[e^{\sigma(W_{t+s}^* - W_s^*)} \,\middle|\, \mathcal{F}_s\right] = e^{\frac{1}{2}\sigma^2 t}.$$

where

W_t^*	the $\mathbb{Q}$-Brownian motion: the old BM plus the drift shift.
$e^{\frac{1}{2}\sigma^2 t}$	moment generating function of $\mathcal{N}(0,t)$ at σ: the hallmark of a standard BM.
$\mathcal{F}_s$	information up to time s (slide writes "$\mid s$"); increments are independent of the past.

Girsanov theorem (general). If W_t is a $\mathbb{P}$-Brownian motion and κ_t satisfies Novikov's condition

$$\mathbb{E}^{\mathbb{P}}\left[\exp\left(\tfrac{1}{2}\int_0^T \kappa_t^2\,\mathrm{d}t\right)\right] < \infty,$$

then there is a measure $\mathbb{Q}$ such that

$$(1)\ \mathbb{Q} \sim \mathbb{P}; \qquad (2)\ \frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}} = \exp\Big(-\int_0^T \kappa_t\,\mathrm{d}W_t - \frac{1}{2}\int_0^T \kappa_t^2\,\mathrm{d}t\Big); \qquad (3)\ W_t^* = W_t + \int_0^t \kappa_u\,\mathrm{d}u \ \text{ is a } \mathbb{Q}\text{-BM}.$$

where

κ_t	the drift shift (Girsanov kernel); may be random but must be adapted.
Novikov condition	guarantees the exponential has expectation 1, so $\mathbb{Q}$ is a genuine probability measure *[Beyond slides]*.
$\mathbb{Q} \sim \mathbb{P}$	equivalent: same null sets.
$-\int \kappa\,\mathrm{d}W$	the stochastic ("$-\kappa x$") part of the density ratio, as in [S3].
$-\frac{1}{2}\int \kappa^2\,\mathrm{d}t$	the normalising ("$-\frac{1}{2}\kappa^2 t$") part.
W_t^*	has zero drift under $\mathbb{Q}$; equivalently $\mathrm{d}W_t = \mathrm{d}W_t^* - \kappa_t\,\mathrm{d}t$.

Recipe (how to use it): write the SDE as $\mathrm{d}X = a\,\mathrm{d}t + b\,\mathrm{d}W$; to get target drift $\tilde{a}$, write $\mathrm{d}X = \tilde{a}\,\mathrm{d}t + b\,(\mathrm{d}W + \kappa\,\mathrm{d}t)$ with $\kappa = \frac{a-\tilde{a}}{b}$; then $\mathrm{d}W^* = \mathrm{d}W + \kappa\,\mathrm{d}t$. Only the drift changes; the volatility b is untouched.

1.5 Girsanov example 1: four expectations [S6]

With W_t a $\mathbb{P}$-BM, $W_t^* = W_t + \kappa t$ a $\mathbb{Q}$-BM (worked answers *[Beyond slides]*):

$$\mathbb{E}^{\mathbb{P}}[W_t] = 0, \qquad \mathbb{E}^{\mathbb{Q}}[W_t] = \mathbb{E}^{\mathbb{Q}}[W_t^* - \kappa t] = -\kappa t, \qquad \mathbb{E}^{\mathbb{Q}}[W_t^*] = 0, \qquad \mathbb{E}^{\mathbb{P}}[W_t^*] = \mathbb{E}^{\mathbb{P}}[W_t + \kappa t] = \kappa t.$$

where

$\mathbb{E}^{\mathbb{Q}}[W_t]$	check via the density: $\mathbb{E}^{\mathbb{P}}\left[W_t\, e^{-\kappa W_t - \frac{1}{2}\kappa^2 t}\right] = -\kappa t$ (differentiate the MGF $e^{\frac{1}{2}\kappa^2 t}$ in κ).
rule	each process is driftless only under its own measure; under the other measure it picks up drift $\pm\kappa$.

1.6 Girsanov example 2: change μ to r [S7]

$$\mathrm{d}X_t = \mu X_t\,\mathrm{d}t + \sigma X_t\,\mathrm{d}W_t = rX_t\,\mathrm{d}t + \sigma X_t\Big(\mathrm{d}W_t + \frac{\mu - r}{\sigma}\,\mathrm{d}t\Big) \implies \mathrm{d}X_t = rX_t\,\mathrm{d}t + \sigma X_t\,\mathrm{d}W_t^*, \quad W_t^* = W_t + \frac{\mu - r}{\sigma}t.$$

where

μ	original ($\mathbb{P}$) drift rate.
r	target drift rate.
$\kappa = \frac{\mu - r}{\sigma}$	the Girsanov kernel; in finance the **market price of risk** (Sharpe ratio) *[Beyond slides]*.
W_t^*	$\mathbb{Q}$-Brownian motion; $\mathbb{Q}$ is the measure with $\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}} = e^{-\kappa W_T - \frac{1}{2}\kappa^2 T}$.

1.7 Before Black–Scholes: actuarial pricing [S8, S10]

- Pre-1973 models (Sprenkle, Boness, Samuelson *[Beyond slides]*) valued an option as the *empirical* expected discounted payoff. The value depended on volatility, but also on which growth rate μ and which risky discount rate to use — a personal forecast [S8, Derman].

$$\text{Call} = S_0\,\Phi\left(\frac{\log\frac{S_0}{K} + (\mu + \frac{\sigma^2}{2})T}{\sigma\sqrt{T}}\right) - Ke^{-fT}\,\Phi\left(\frac{\log\frac{S_0}{K} + (\mu - \frac{\sigma^2}{2})T}{\sigma\sqrt{T}}\right) \quad \text{[S10]}.$$

where

μ	the stock's expected return (unknown, must be forecast).
f	the discount rate chosen for the option's risk ("personal").

⚠ The slide formula drops a factor on the first term. Discounting $\mathbb{E}^{\mathbb{P}}[(S_T - K)^+]$ at rate f when S grows at μ gives $S_0 e^{(\mu-f)T}\Phi(d_1^{\mu}) - Ke^{-fT}\Phi(d_2^{\mu})$. The slide version is exact only when $f = \mu$. With $\mu = f = r$ it becomes Black–Scholes.

1.8 Black–Scholes (1971–73) and the Nobel Prize [S9–S11]

$$\text{Call} = S_0\,\Phi\left(\frac{\log\frac{S_0}{K} + (r + \frac{\sigma^2}{2})T}{\sigma\sqrt{T}}\right) - Ke^{-rT}\,\Phi\left(\frac{\log\frac{S_0}{K} + (r - \frac{\sigma^2}{2})T}{\sigma\sqrt{T}}\right).$$

where

r	riskless rate: replaces *both* μ (growth) and f (discount).
σ	future volatility of the stock: the only quantity still requiring a view.

- Hedge the stock risk out of the option. The hedged portfolio is riskless, so it must earn the riskless rate. The formula has the same shape as the actuarial one, but every growth and discount rate is r [S9].
- The value does **not** depend on the stock's expected return, because that risk has been hedged away [S9].
- 1997 Nobel Prize to Merton and Scholes (Black died in 1995): the method paved the way for valuation in many areas, new instruments, and more efficient risk management [S11].

1.9 Black–Scholes market and assumptions [S12]

$$\begin{cases} \mathrm{d}S_t = \mu S_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t \\ \mathrm{d}B_t = rB_t\,\mathrm{d}t \end{cases}$$

where

S_t	risky asset (stock).
B_t	risk-free asset (bond / money-market account), $B_t = B_0 e^{rt}$.

Assumptions: (1) underlying is lognormal with constant mean and variance; (2) r constant; (3) no dividends during the option's life; (4) short selling permitted; (5) no risk-free arbitrage; (6) continuous-time trading; (7) no transaction costs, taxes or trading limits.

1.10 Choice of numeraire [S13]

Under the risk-neutral measure the stock grows at r:

$$\mathbb{E}_t^*[S_{t+\Delta t}] = S_t e^{r\Delta t} \iff \frac{S_t}{e^{rt}} = \mathbb{E}_t^*\left[\frac{S_{t+\Delta t}}{e^{r(t+\Delta t)}}\right].$$

where

$\mathbb{E}_t^*$	expectation under the risk-neutral measure $\mathbb{Q}^*$, given information at t.
e^{rt}	the **numeraire**: the unit in which prices are measured (here the money-market account).
S_t/e^{rt}	the price ratio: a $\mathbb{Q}^*$-martingale — its best forecast is today's value.

- The only requirement for a numeraire is that it is **strictly positive** at all times [S13].
- The money-market account is the popular choice (Harrison–Kreps 1979), but the choice is arbitrary [S13, S15]. *[Beyond slides]* Each numeraire N_t has its own martingale measure $\mathbb{Q}^N$ with $\frac{V_t}{N_t} = \mathbb{E}_t^N\left[\frac{V_T}{N_T}\right]$ (e.g. the zero-coupon bond gives the T-forward measure).

1.11 Equivalent martingale measure: key concepts [S14–S15]

- In a **complete market** every derivative is attainable (replicable). A perfectly hedged derivative loses its randomness and behaves like a riskless bond [S14].
- So real-world probabilities do not enter risk-neutral valuation; the hedged portfolio grows at r, hence (hedged portfolio)$/B_t$ is a martingale [S14].
- Two measures are **equivalent** if they agree on what is possible and impossible [S14]. So an arbitrage in one measure is an arbitrage in every equivalent measure [S15].
- If the risk-neutral price is arbitrage-free, it is arbitrage-free in the real world [S15].
- If prices divided by a numeraire are martingales, there is no arbitrage. Under the associated measure, the discounted option price is a martingale too, giving its present value [S15]. *[Beyond slides]* This is the First Fundamental Theorem of Asset Pricing; uniqueness of the EMM ⇔ market completeness is the Second.

1.12 EMM application: which parameter is hardest? [S16]

In $\mathrm{d}B_t = rB_t\,\mathrm{d}t$, $\mathrm{d}S_t = \mu S_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t$ (W_t a $\mathbb{P}$-BM), the answer *[Beyond slides]* is $\boldsymbol{\mu}$. r is observed in the market; σ can be estimated well from high-frequency data (or implied from options). The standard error of an estimated mean return is $\sigma/\sqrt{T_{\text{years}}}$, which depends only on the *length* of the history, so even decades of data leave μ very uncertain. EMM pricing removes μ entirely.

1.13 Relative price and the martingale measure [S17]

With $B_0 = 1$, define $X_t = S_t/B_t = f(S_t, B_t)$. Itô (*[Beyond slides]* working: $f_S = 1/B$, $f_B = -S/B^2$, $f_{SS} = 0$, and $\mathrm{d}B$ has no $\mathrm{d}W$):

$$\mathrm{d}X_t = \frac{\mathrm{d}S_t}{B_t} - \frac{S_t}{B_t^2}\,\mathrm{d}B_t = (\mu - r)X_t\,\mathrm{d}t + \sigma X_t\,\mathrm{d}W_t.$$

Girsanov with $\kappa = \frac{\mu - r}{\sigma}$: $\mathrm{d}W_t^* = \mathrm{d}W_t + \frac{\mu - r}{\sigma}\mathrm{d}t$, so

$$\mathrm{d}X_t = (\mu - r)X_t\,\mathrm{d}t + \sigma X_t\Big(\mathrm{d}W_t^* - \frac{\mu - r}{\sigma}\mathrm{d}t\Big) = \sigma X_t\,\mathrm{d}W_t^*.$$

where

X_t	stock price in units of the money-market account (discounted price).
$(\mu - r)X_t$	real-world drift of the discounted price: the excess return.
W_t^*	$\mathbb{Q}^*$-Brownian motion; "$*$" signals the numeraire is B_t (most common choice).
$\sigma X_t\,\mathrm{d}W_t^*$	no drift: X_t is a $\mathbb{Q}^*$-martingale.

This is the **only** measure that makes the relative price a martingale [S18] (one stock, one BM ⇒ unique EMM ⇒ complete market *[Beyond slides]*).

1.14 Stock under $\mathbb{Q}^*$ [S18]

$$\mathrm{d}S_t = \mu S_t\,\mathrm{d}t + \sigma S_t\Big(\mathrm{d}W_t^* - \frac{\mu - r}{\sigma}\mathrm{d}t\Big) = rS_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t^* \implies S_T = S_0\exp\Big[\Big(r - \frac{1}{2}\sigma^2\Big)T + \sigma W_T^*\Big].$$

where

r	replaces μ: under the EMM the stock's drift is irrelevant.
$-\frac{1}{2}\sigma^2 T$	Itô correction from solving GBM (Session 5).
$W_T^* \sim \mathcal{N}(0,T)$	under $\mathbb{Q}^*$; write $W_T^* = \sqrt{T}\,x$, $x \sim \mathcal{N}(0,1)$.

1.15 Martingale pricing of a European call [S19]

For $V_T = (S_T - K)^+$:

$$\frac{V_0}{B_0} = \mathbb{E}^*\left[\frac{V_T}{B_T}\right] = \mathbb{E}^*\left[\frac{(S_T-K)^+}{B_T}\right] = \frac{e^{-rT}}{\sqrt{2\pi}}\int_{-\infty}^{\infty}\left[S_0 e^{(r-\frac{1}{2}\sigma^2)T+\sigma\sqrt{T}x} - K\right]^+ e^{-\frac{x^2}{2}}\,\mathrm{d}x$$

$$= S_0\Phi(d_1) - Ke^{-rT}\Phi(d_1-\sigma\sqrt{T}), \qquad d_1 = \frac{\log\frac{S_0}{K} + (r+\frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}.$$

where

V_0/B_0	today's discounted value ($B_0 = 1$).
$\mathbb{E}^*[V_T/B_T]$	martingale pricing theorem: the discounted price is a $\mathbb{Q}^*$-martingale.
$B_T = e^{rT}$	deterministic, so it comes out of the expectation as e^{-rT}.
$[\cdot]^+$	positive only for $x > -d_2$ where $d_2 = d_1 - \sigma\sqrt{T}$; split the integral there.
$S_0\Phi(d_1)$	from the stock term: complete the square $(x - \sigma\sqrt{T})$ *[Beyond slides].*
$Ke^{-rT}\Phi(d_2)$	from the strike term: $\mathbb{Q}^*(S_T > K) = \Phi(d_2)$.

1.16 Implementation *[Beyond slides]* (tested)

```
import numpy as np
from scipy.stats import norm
rng = np.random.default_rng(0)
def bs_call(S0, K, r, sigma, T):
    d1 = (np.log(S0/K) + (r + 0.5*sigma**2)*T) / (sigma*np.sqrt(T))
    return S0*norm.cdf(d1) - K*np.exp(-r*T)*norm.cdf(d1 - sigma*np.sqrt(T))
# 1) Girsanov: reweight P-paths by dQ/dP = exp(-kappa W_t - kappa^2 t/2)
kappa, t, n = 0.5, 2.0, 2_000_000
W = np.sqrt(t)*rng.standard_normal(n)             # P-Brownian motion at t
Z = np.exp(-kappa*W - 0.5*kappa**2*t)             # Radon-Nikodym derivative
print(round(Z.mean(), 3))                         # 0.999  ~ E_P[dQ/dP] = 1
print(round((W*Z).mean(), 3))                     # -0.998 ~ E_Q[W_t] = -kappa t = -1
print(round(((W + kappa*t)*Z).mean(), 3))         # 0.002  ~ E_Q[W*_t] = 0
# 2) Martingale pricing: simulate S_T under Q*, discount by B_T (mu never used)
S0, K, r, mu, sigma, T = 100, 105, 0.05, 0.12, 0.2, 1.0
Ws = np.sqrt(T)*rng.standard_normal(n)
ST = S0*np.exp((r - 0.5*sigma**2)*T + sigma*Ws)
print(round(np.exp(-r*T)*np.maximum(ST - K, 0).mean(), 3),
      round(bs_call(S0, K, r, sigma, T), 3))      # 8.006 (MC)  8.021 (closed form)
print(round(np.exp(-r*T)*ST.mean(), 3))           # 99.975 ~ S_0: S/B is a Q*-martingale
```

2 Comparison tables

2.1 Real-world vs risk-neutral

	Real world $\mathbb{P}$ [S12, S16]	**Risk-neutral** $\mathbb{Q}^*$ [S17–S18]
Stock drift	μ (unknown)	r
Brownian motion	W_t	$W_t^* = W_t + \frac{\mu-r}{\sigma}t$
Volatility	σ	σ (unchanged by Girsanov)
S_t/B_t	drift $(\mu - r)$: not a martingale	martingale
Use	forecasting, risk management	pricing derivatives
Discount rate for options	unknown, risk-dependent	r

2.2 Discrete vs continuous measure change [S3]

	Discrete (binomial tree)	**Continuous (Brownian)**
Radon–Nikodym	ratio of probabilities, e.g. q/p, $(1-q)/(1-p)$	ratio of densities $f/\tilde{f}$
Form	product of node ratios along a path	$e^{-\kappa W_T - \frac{1}{2}\kappa^2 T}$
Equivalence	$0 < p, q < 1$	density ratio > 0
Effect	changes up/down probabilities	changes the drift only

2.3 Actuarial vs Black–Scholes pricing [S8–S10]

	Actuarial (pre-1973)	**Black–Scholes**
Growth rate of S	μ (forecast)	r
Discount rate	f (risk-adjusted, personal)	r
Inputs	$S_0, K, T, \sigma, \mu, f$	S_0, K, T, σ, r
Justification	expected value	hedging / no arbitrage

3 Master formula sheet

Each line repeats the key symbols so the sheet can be used on its own.

Radon–Nikodym derivative [S3–S4]

- $\frac{\mathrm{d}\mathbb{P}}{\mathrm{d}\tilde{\mathbb{P}}} = \frac{f(x)}{\tilde{f}(x)} = e^{-\kappa x - \frac{1}{2}\kappa^2 t} > 0$; $\mathbb{E}^{\tilde{\mathbb{P}}}\big[\frac{\mathrm{d}\mathbb{P}}{\mathrm{d}\tilde{\mathbb{P}}}\big] = 1$; $\mathbb{E}^{\mathbb{P}}[g] = \mathbb{E}^{\tilde{\mathbb{P}}}\big[g\frac{\mathrm{d}\mathbb{P}}{\mathrm{d}\tilde{\mathbb{P}}}\big]$.
  where f = density of $\mathcal{N}(-\kappa t, t)$, $\tilde{f}$ = density of $\mathcal{N}(0, t)$.

Girsanov [S5]

- $\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}} = \exp\big(-\int_0^T \kappa_t\,\mathrm{d}W_t - \frac{1}{2}\int_0^T \kappa_t^2\,\mathrm{d}t\big)$, $W_t^* = W_t + \int_0^t \kappa_u\,\mathrm{d}u$ is a $\mathbb{Q}$-BM.
  where κ = drift shift; Novikov: $\mathbb{E}[e^{\frac{1}{2}\int \kappa^2 \mathrm{d}t}] < \infty$.
- Drift change: $\mathrm{d}X = a\,\mathrm{d}t + b\,\mathrm{d}W \to \tilde{a}\,\mathrm{d}t + b\,\mathrm{d}W^*$ with $\kappa = (a - \tilde{a})/b$.

Black–Scholes under the EMM [S17–S19]

- $\kappa = \frac{\mu - r}{\sigma}$; $\mathrm{d}(S/B) = \sigma(S/B)\,\mathrm{d}W^*$; $\mathrm{d}S = rS\,\mathrm{d}t + \sigma S\,\mathrm{d}W^*$; $S_T = S_0 e^{(r - \frac{1}{2}\sigma^2)T + \sigma W_T^*}$.
- $\frac{V_0}{B_0} = \mathbb{E}^*\big[\frac{V_T}{B_T}\big]$; call $= S_0\Phi(d_1) - Ke^{-rT}\Phi(d_2)$, $d_{1,2} = \frac{\log(S_0/K) + (r \pm \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$.

Numeraire [S13]

- $\frac{S_t}{e^{rt}} = \mathbb{E}_t^*\big[\frac{S_{t+\Delta t}}{e^{r(t+\Delta t)}}\big]$; numeraire must be strictly positive.

4 Glossary

- **Complete market:** every derivative can be replicated [S14].
- **Equivalent measures:** agree on which events have probability zero [S14].
- **Equivalent martingale measure (EMM):** an equivalent measure under which prices divided by the numeraire are martingales [S14–S15].
- **Girsanov's theorem:** exponential change of measure that shifts the drift of a BM [S5].
- **Likelihood ratio / Radon–Nikodym derivative:** $\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}$, the reweighting factor [S3].
- **Market price of risk:** $\kappa = (\mu - r)/\sigma$ *[Beyond slides].*
- **Martingale pricing theorem:** $V_0/N_0 = \mathbb{E}^N[V_T/N_T]$ [S19].
- **Money-market account:** $B_t = B_0 e^{rt}$ [S15].
- **Novikov condition:** $\mathbb{E}[e^{\frac{1}{2}\int_0^T \kappa_t^2 \mathrm{d}t}] < \infty$ [S5].
- **Numeraire:** strictly positive asset used as the unit of account [S13].
- **Relative price:** $X_t = S_t/B_t$ [S17].

5 Self-test

Questions

Q1 ([S2–S3]). Derive the likelihood ratio between $\mathcal{N}(-\kappa t, t)$ and $\mathcal{N}(0, t)$.
Q2 ([S4]). State three properties of the Radon–Nikodym derivative. Why are the two measures equivalent?
Q3 ([S5]). State Girsanov's theorem.
Q4 ([S6]). With $W_t^* = W_t + \kappa t$, compute $\mathbb{E}^{\mathbb{P}}[W_t]$, $\mathbb{E}^{\mathbb{Q}}[W_t]$, $\mathbb{E}^{\mathbb{Q}}[W_t^*]$, $\mathbb{E}^{\mathbb{P}}[W_t^*]$.
Q5 ([S7]). Change the measure so that $\mathrm{d}X = \mu X \mathrm{d}t + \sigma X \mathrm{d}W$ has drift r. What is $\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}$?
Q6 ([S8–S10]). Why does the Black–Scholes price not depend on μ?
Q7 ([S13]). What is a numeraire and what is the only requirement?
Q8 ([S14–S15]). Why is a price that is arbitrage-free under $\mathbb{Q}^*$ also arbitrage-free under $\mathbb{P}$?
Q9 ([S17]). Derive $\mathrm{d}(S/B)$ and find the measure that makes it a martingale.
Q10 ([S18–S19]). Derive the Black–Scholes call price by martingale pricing.

Solutions

S1. Ratio of densities: $\exp[-\frac{(x+\kappa t)^2 - x^2}{2t}] = e^{-\kappa x - \frac{1}{2}\kappa^2 t}$.
S2. Strictly positive; expectation 1 under $\tilde{\mathbb{P}}$; $\mathbb{E}^{\mathbb{P}}[g] = \mathbb{E}^{\tilde{\mathbb{P}}}[g\,\mathrm{d}\mathbb{P}/\mathrm{d}\tilde{\mathbb{P}}]$. Both densities are positive everywhere, so they share null sets.
S3. See §1.4: Novikov $\Rightarrow \mathbb{Q} \sim \mathbb{P}$, density $e^{-\int \kappa \mathrm{d}W - \frac{1}{2}\int \kappa^2 \mathrm{d}t}$, $W^* = W + \int \kappa\,\mathrm{d}u$ is a $\mathbb{Q}$-BM.
S4. 0, $-\kappa t$, 0, κt.
S5. $\kappa = \frac{\mu - r}{\sigma}$, $W^* = W + \kappa t$, $\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}} = e^{-\kappa W_T - \frac{1}{2}\kappa^2 T}$; $\mathrm{d}X = rX\mathrm{d}t + \sigma X \mathrm{d}W^*$.
S6. The stock risk is hedged away; the hedged portfolio is riskless and earns r. Formally, Girsanov changes only the drift, and pricing under $\mathbb{Q}^*$ uses drift r.
S7. The unit in which prices are expressed; it must be strictly positive at all times.
S8. Equivalent measures agree on which events are possible, and an arbitrage is defined only through possible/impossible events ($V_0 \le 0$, $V_T \ge 0$ a.s., $\mathbb{P}(V_T > 0) > 0$).
S9. $\mathrm{d}X = (\mu - r)X\mathrm{d}t + \sigma X \mathrm{d}W$; Girsanov with $\kappa = (\mu - r)/\sigma$ gives $\mathrm{d}X = \sigma X \mathrm{d}W^*$.
S10. See §1.15.

Appendix A. Slide coverage checklist

Slide	Content	Section
S1	Title	header
S2	Densities with and without drift	1.1
S3	Likelihood ratio, Radon–Nikodym derivative	1.2
S4	Properties; change of measure; equivalence (question)	1.3
S5	Girsanov theorem	1.4
S6	Example: four expectations	1.5
S7	Example: change drift $\mu \to r$	1.6
S8	Derman: before Black–Scholes	1.7
S9	Derman: Black–Scholes hedging argument	1.8
S10	Actuarial vs Black–Scholes formulas	1.7, 1.8
S11	1997 Nobel citation	1.8
S12	Black–Scholes market and assumptions	1.9
S13	Choice of numeraire	1.10
S14	EMM key concepts (complete market, equivalence)	1.11
S15	EMM: arbitrage across measures; Harrison–Kreps	1.11
S16	Which parameter is hardest to estimate (question)	1.12
S17	Relative price, Girsanov, $\mathrm{d}X = \sigma X \mathrm{d}W^*$	1.13
S18	Stock under $\mathbb{Q}^*$ and its solution	1.14
S19	Martingale pricing of the call	1.15

Appendix B. Flag summary (⚠)

1. [S2–S5] The roles of the measures flip: on [S2–S4] $\mathbb{P}$ carries the drift $-\kappa$, on [S5] $\mathbb{Q}$ does. The density formula is the same once you track which is which.
2. [S10] The "before Black–Scholes" formula omits $e^{(\mu-f)T}$ on the S_0 term; it is exact only when $f = \mu$.
3. [S5] The conditional expectation is written "$\mid s$"; it means conditioning on the information $\mathcal{F}_s$ up to time s.
4. [S4, S6, S16] Questions posed on the slides (equivalence, the four expectations, the hardest parameter) are answered here with my own working.